Inward Pointing Vectors on Manifolds with Corners

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06.11.2026

Model with Corners

We will focus on real manifolds.
Let E be a normed space over $\mathbb{R}$ and H a topological space.

Definition: Model with Corners

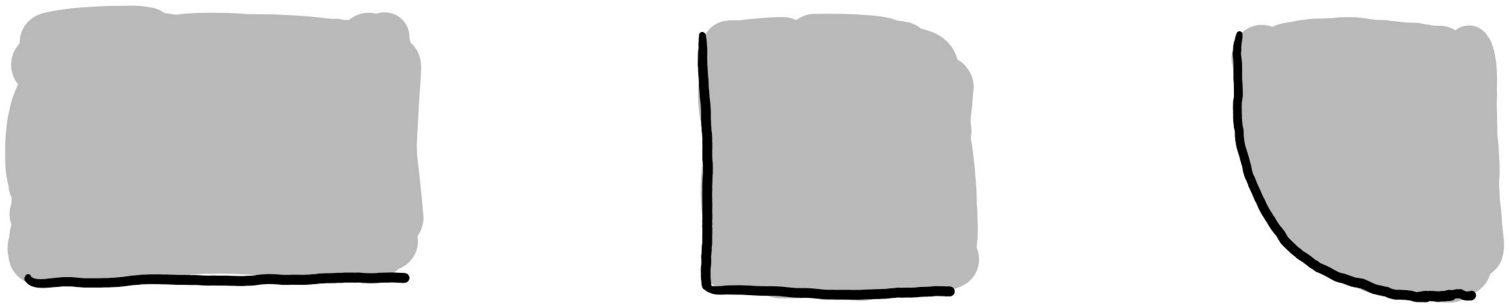
A *model with corners* from E to H is a partial equivalence $I : H \to E$ such that

(i) the forward and backward map are continuous,

(ii) then source is the entirety of H, and

(iii) the range is convex and has nonempty interior.

Intuition: Model with Corners



Different examples of models with corners

Topological Manifold with Corners

Definition: Topological Manifold with Corners

A topological space M is a *topological n-manifold* if

(i) M is Hausdorff.

(ii) M is second-countable, i.e. M has a countable base.

(iii) M is locally H, i.e. for every point p there is a neighbourhood of p that is homeomorphic to an open set in H.

Definition: Chart

Let M be a topological manifold with corners. Let U be an open subset of M and $\varphi\colon U \to H$ a homeomorphism onto an open subset of H. Then we call (U, φ) a chart of M.

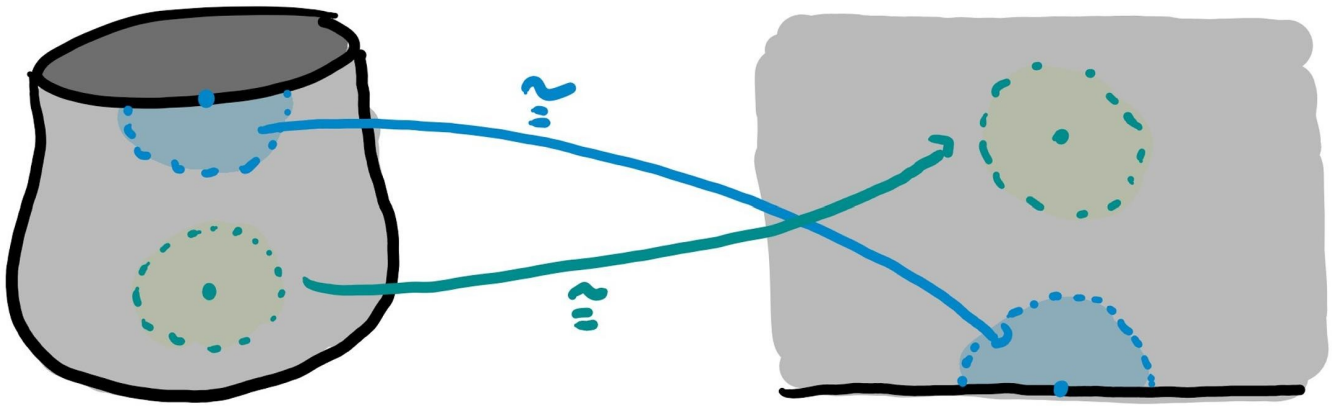
Boundary of a Topological Manifold

Definition: Boundary

Let M be a topological manifold with boundary.

(i) A point $p \in M$ is a *boundary point* if it is in the domain of a *boundary chart*, i.e. a chart (U, φ) with $I(\varphi(p)) \in \partial(I(H))$.

(ii) A point $p \in M$ is an *interior point* if it is in the domain of an *interior chart*, i.e. a chart (U, φ) with $I(\varphi(U)) \cap \partial(I(H)) = \varnothing$.



Smooth Manifold with Corners

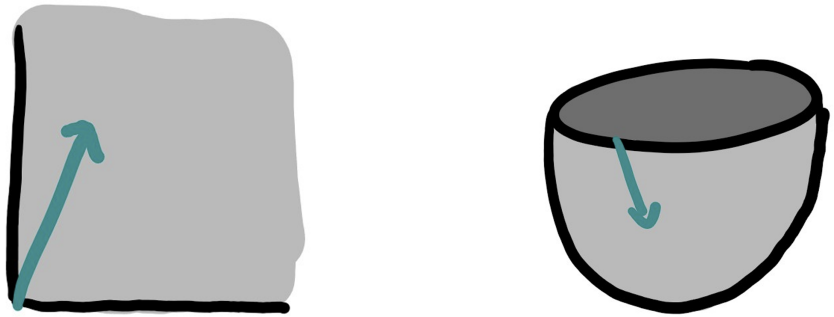
Definition: Smooth Atlas

Let M be a topological manifold. Let $\mathcal{A}$ be a collection of charts whose domains cover M. $\mathcal{A}$ is a smooth atlas for M if any two charts (φ, U) and (ψ, V) are *smoothly compatible*, i.e. either $U \cap V$ is empty or $I \circ \psi \circ \varphi^{-1} \circ I^{-1} \colon I(\varphi(U \cap V)) \to \psi I((U \cap V))$ is smooth in the ordinary sense.

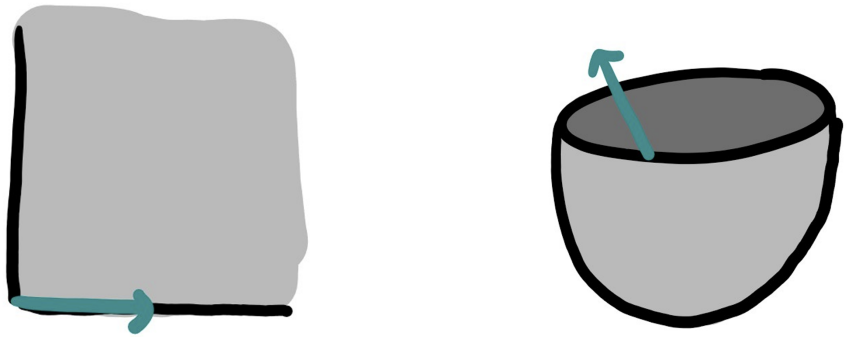
Definition: Smooth Manifold

A pair $(M, \mathcal{A})$ of a topological manifold M and a smooth altas $\mathcal{A}$ on M, which is maximal with respect to inclusion, is called a *smooth manifold*.

Intuition: Inward pointing vectors



Example of inward pointing vectors

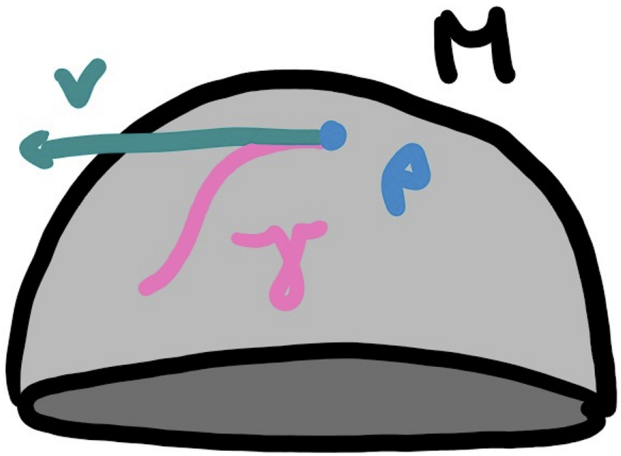


Different examples of models with corners

Definition on Manifold without Corners

Inward pointing [Lee06]

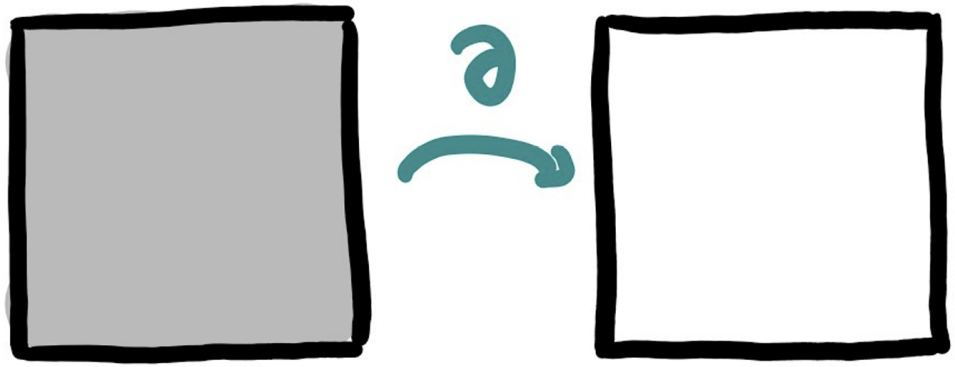
Let M be a smooth manifold with boundary without corners. Let $p \in M$. Then $v \in T_pM \setminus T_p\partial M$ is inward pointing if and only if there is an $\varepsilon > 0$ and a smooth curve $\gamma\colon [0, \varepsilon) \to M$ such that $\gamma(0) = p$ and $\gamma'(0) = v$.



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Options for definition

Definition: (weakly) inward pointing/realisable

We call a vector that is inward pointing or tangent to the boundary *realisable* or *weakly inward pointing*.

We have the following two options for the definition of inward-pointing vectors:

(A) intrinsically exclude tangent vectors, or

(B) define realisable vectors and then exclude tangent vectors

Option A: attempt 1

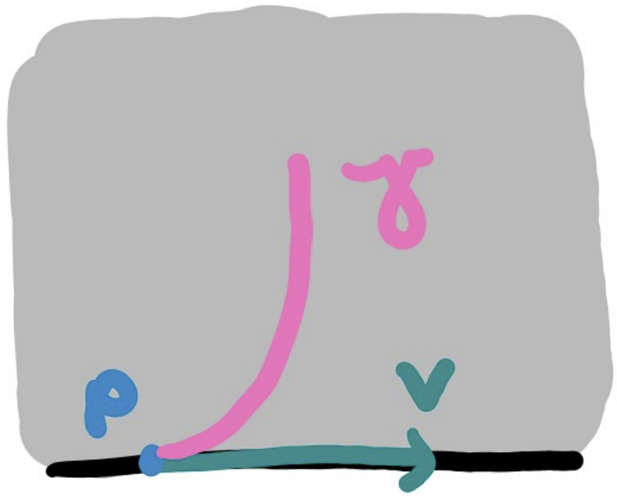
Attempt 1: our idea inspired by [Lee06]

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Option A: attempt 2

Attempt 2: adapted from [Mel96]

Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is inward pointing if and only if $D_v f(p) \geq 0$ for every non-negative $f \in C^\infty(M)$ that is zero on ∂M.

This definition is clearly designed to mean weakly inward-pointing.

Option A: attempt 3

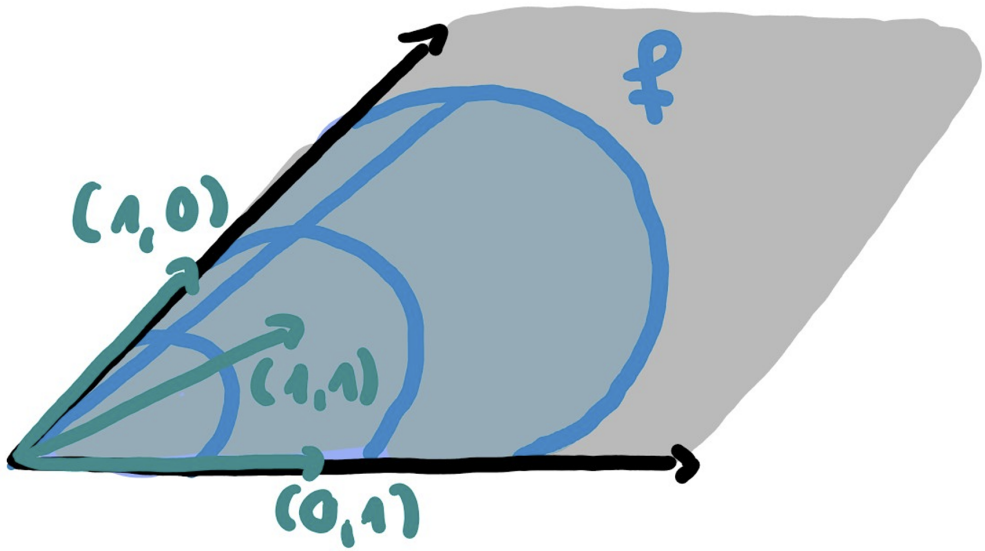
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Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is inward pointing if and only if there is a non-negative $f \in C^\infty(M)$ that is zero on ∂M such that $\mathrm{D}_v f(p) > 0$.

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Option A: attempt 4

Attempt 4: e.g. [Joy09]

Let M be a smooth manifold with corners over the model with corners $\mathbb{R}^n_k = [0,\infty)^k \times \mathbb{R}^{n-k}$. Let $p \in M$ be a corner point of codimension k, i.e. let there be a chart (U, φ) with $U \subseteq \mathbb{R}^n_k$ such that $0 \in U$ and $\varphi(U) = p$. Then a vector $v \in T_pM$ is inward pointing if and only if it is in $\mathrm{d}\varphi_0((0,\infty)^k \times \mathbb{R}^{n-k})$ where $\mathrm{d}\varphi_0 \colon \mathbb{R}^n = T_0\mathbb{R}^n_k \to T_pM$.

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Defining inward-pointing from realisability

Definition: inward-pointing, [Amr21]

Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is inward pointing if and only if it is in the interior of the set of realisable vectors.

Conditions for a definition inward-pointing

A sensible definition of realisability needs to fulfil these two conditions:

(i) The definition needs to be preserved by local diffeomorphisms (in particular it should be preserved by smooth charts).

(ii) The definition should reduce to a sensible notion on the model space that is easily verifiable in the cases of the typical examples (e.g. upper half space in $\mathbb{R}^n$).

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Sensible notion in normed spaces: tangent cones

Definition: contingent tangent cone, [AF90]

Let E be a normed vector space, $K \subset E$ and $x \in \overline{E}$. Then the *contingent tangent cone* to K at x is the set of all $v \in E$ such that there is a sequence of scalars $c\colon \mathbb{N} \to \mathbb{R}_{\geq}$ and a sequence of vectors $d\colon \mathbb{N} \to E$ such that

(i) $\lim_{n\to\infty} d_n = 0$,

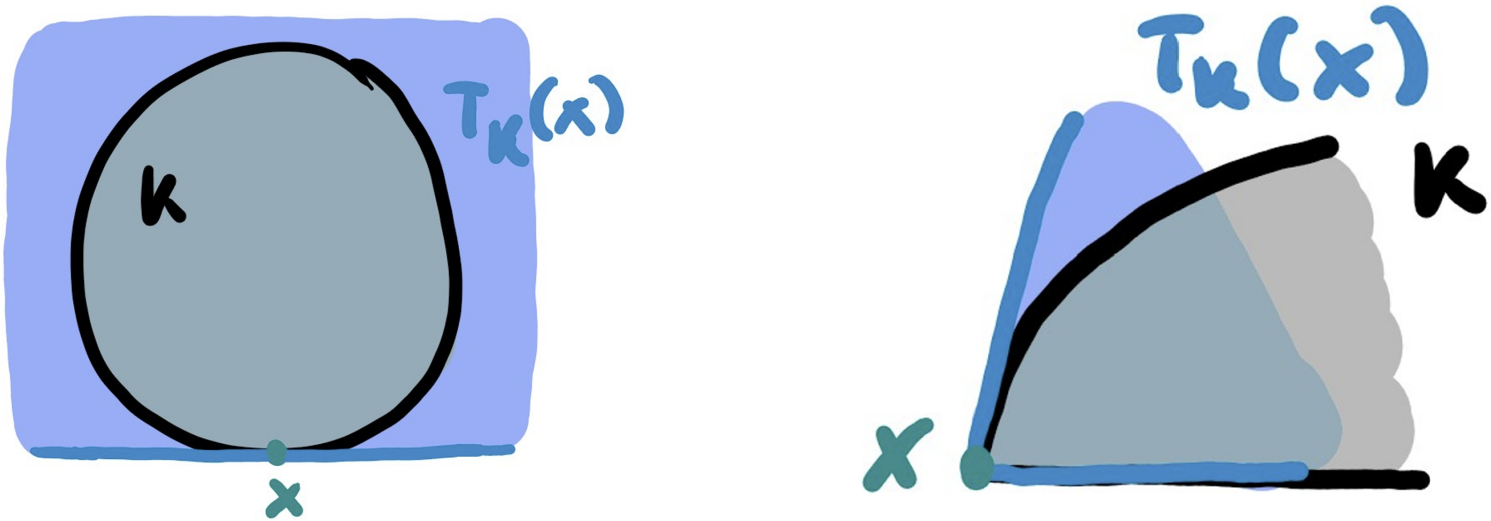
(ii) $x + d_n \in K$ for all $n \in \mathbb{N}$, and

(iii) $\lim_{n\to\infty} c_n \cdot d_n = v$.

We denote this tangent cone by $T_K(x)$.

This is in Mathlib as `posTangentConeAt`. For convex sets this notion agrees with other types of tangent cones such as the Clarke tangent cone.

Tangent Cone: Intuition



Examples of tangent cones

Option B: attempt 1

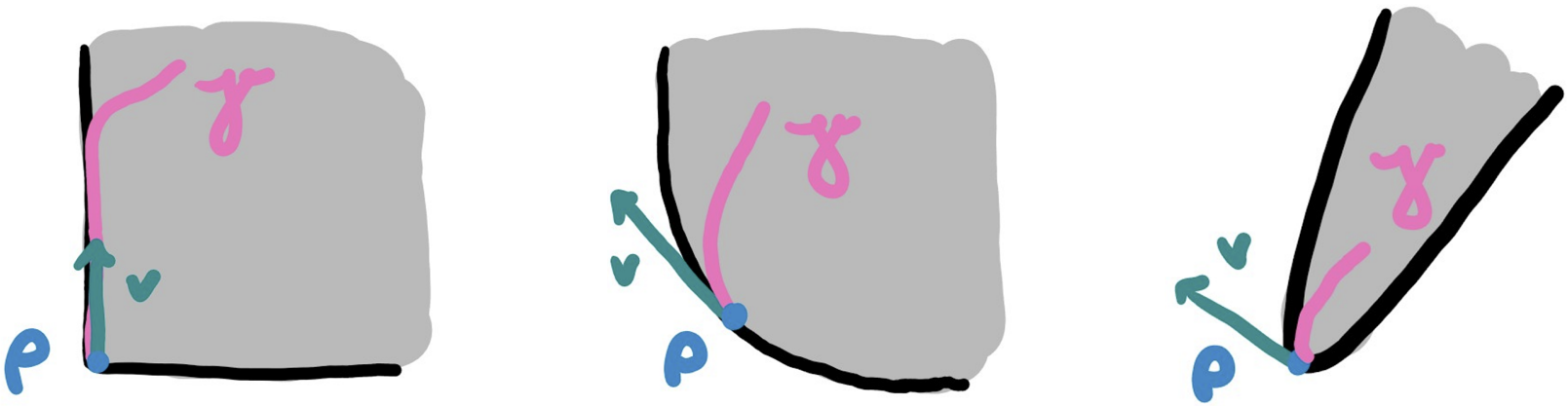
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Realisability examples

Option B: attempt 1

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Requiring γ to be continuous instead of smooth would work better.

Option B: final pick

Definition: Realisability, [RW98]

Let M be a smooth manifold with corners. Let $p \in M$. Then $v \in T_pM$ is *realisable* if and only if there is a function $\gamma\colon [0,\infty) \to M$ that is differentiable at 0 and fulfils $\gamma(0) = p$ and $\gamma'(0) = v$.

```
def IsRealizable {M H E : Type*} [TopologicalSpace H]
    [TopologicalSpace M] [ChartedSpace H M]
    [NormedAddCommGroup E] [NormedSpace ℝ E]
    {I : ModelWithCorners ℝ E H} {p : M}
    (v : TangentSpace I p) : Prop :=
  ∃ (γ : ℝ → M) (_ : MDiffAt[Ici 0] γ 0), γ 0 = p ∧
    mfderiv[Ici 0] γ (0 : ℝ) 1 = v
```

Invariance under charts

Lemma: invariance under charts

Let $f : M \to M'$ be a local diffeomorphism between manifolds with corners. Let $p \in M$ and $v \in T_pM$. Then v is realisable if and only if $\mathrm{D}_v f(p)$ is realisable.

Equivalence to tanget cone I

Definition: derivable cone

Let E be a normed space over $\mathbb{R}$, let $C \subseteq E$ be a set and let $p \in E$ be a point. We call a vector v *derivable* in C at p if there is and $\varepsilon > 0$ and a function $\gamma\colon [0, \varepsilon) \to E$ that is differentiable at 0 with $\gamma([0, \varepsilon)) \subseteq C$, $\varepsilon(0) = p$ and $\varepsilon'(0) = v$. The set of all derivable vectors in C at p is called the *derivable cone* to C at p.

```
def derivableWithinAt (𝕜 : Type*) {E : Type*}
    [NontriviallyNormedField 𝕜]
    [NormedAddCommGroup E] [NormedSpace 𝕜 E]
    [LinearOrder 𝕜] (s : Set E) (p : E)
    (v : E) : Prop :=
  ∃ (γ : 𝕜 → E)
      (_ : DifferentiableWithinAt 𝕜 γ (Ici 0) 0),
      γ 0 = p ∧ derivWithin γ (Ici 0) (0 : 𝕜) = v ∧
      ∀ᶠ (x : 𝕜) in 𝓝[≥] 0, γ x ∈ s
```

Equivalence to tanget cone II

Lemma: derivable and tangent cones agree

Let E be a normed space over $\mathbb{R}$, let $C \subseteq E$ be a convex set and let $p \in C$ be a point. Then the derivable cone and the tangent cone to C at p agree.

Proof steps:

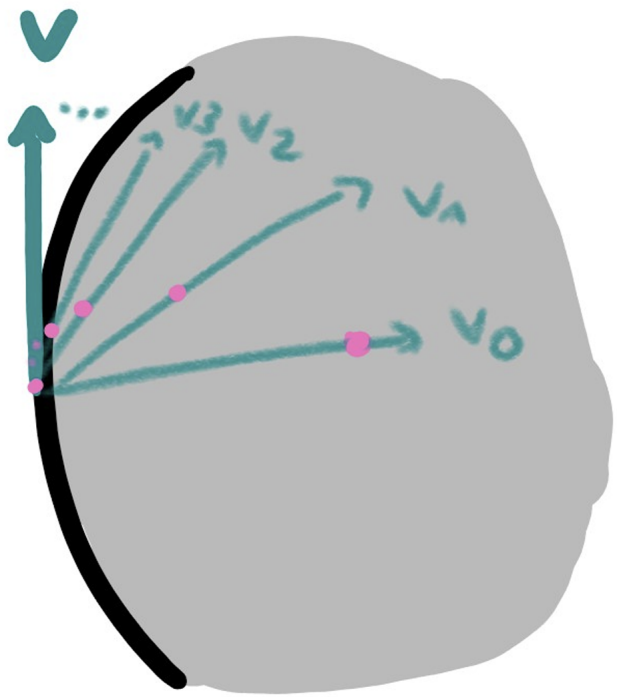
1. Show that the tangent cone is the closure of feasible cone.
2. Show that the derivable cone is closed.
3. Show that the derivable cone equals the closure of the feasible cone.

Equivalence to tangent cone: step 1

Definition: feasible cone

The set of all vectors v such that there is an $r > 0$ such that $r \cdot v + p \in C$ is called the *feasible cone* to C at p.

Equivalence to tangent cone: step 2



The derivable cone is closed

References I

[AF90] Jean-Pierre Aubin and Hélène Frankowska. *Set-valued analysis*. Systems & control 2. Boston: Birkhaeuser, 1990.

[Amr21] Amr (https://math.stackexchange.com/users/29267/amr). *Definition of an inward-pointing tangent vector for smooth manifolds with corners*. Mathematics Stack Exchange. 2021.

[Joy09] Dominic Joyce. "On manifolds with corners". In: *Advances in Geometric Analysis* (Oct. 2009).

References II

[Lee06] John M. Lee. *Introduction to smooth manifolds*. Graduate texts in mathematics 218. New York: Springer, 2006.

[Mel96] Richard Melrose. *Differential analysis on manifolds with corners*. Sept. 1996.

[RW98] Ralph Tyrrell Rockafellar and Roger J.-B. Wets. *Variational analysis*. Grundlehren der mathematischen Wissenschaften 317. Springer, 1998.